

1.3

Logical Equivalence and the Laws of Logic

In this module we define the logical equivalence of wffs, state the principle of duality and prove the laws of logic.

Logical Equivalence

Let a and b be two wffs and let $p_1, p_2, \dots, p_n$ be all the propositional variables occurring in a and b .

We say that a and b are **equivalent** (or **logically equivalent**) if the truth values of a and b are equal for each of the 2^n possible combinations of truth values assigned to $p_1, p_2, \dots, p_n$.

Theorem 1:

a and b are equivalent if and only if $a \leftrightarrow b$ is a tautology.

Proof: Suppose a and b are equivalent. That is, a and b have the same truth value for each of the 2^n possible combinations of truth values assigned to proposition variables $p_1, p_2, \dots, p_n$ and therefore, $a \leftrightarrow b$ is T for each of the 2^n combinations of truth values. Thus, $a \leftrightarrow b$ is a tautology. Conversely, if $a \leftrightarrow b$ is a tautology, then a and b have the same truth value for each of the 2^n combinations of truth values assigned to its n propositional variables. Thus a and b are equivalent.

Representation: We represent the equivalence of a and b by $a \Leftrightarrow b$ or $a \equiv b$, read as a is equivalent to b .

Note:

1. The symbol $\Leftrightarrow$ (or $\equiv$) is not a connective in the object language but a symbol in the meta language and it is a relation.
2. Note that $a \equiv b$ is same as $b \equiv a$ (i. e., a and b have the same truth values for each assignment of the truth values for the proposition variables). Thus, equivalence of wffs is a *Symmetric relation*. Further, if $a \equiv b$ and $b \equiv c$ then $a \equiv c$. That is, the equivalence of wffs is a *Transitive relation*.

Duality Law (Principle of Duality)

We shall consider wffs which contain the connectives $\wedge, \vee$ and $\sim$. (There is no loss of generality in restricting our consideration to these three connectives since any formula containing any other connectives can be replaced by an equivalent formula containing only these three connectives). We introduce two special variables T_0 and F_0 denoting a tautology and contradiction respectively.

Dual of a formula

Let a be a formula containing logical connectives $\wedge, \vee, \sim$ and special variables T_0 and F_0 . The **dual** of a , denoted by a^* , is the formula obtained from a by replacing each occurrence of $\wedge$ and $\vee$ by $\vee$ and $\wedge$ respectively and each occurrence of T_0 and F_0 by F_0 and T_0 respectively.

The connectives $\wedge$ and $\vee$ are also called *duals* of each other.

Example: The duals of $p \vee \sim p$, $p \vee T_0$ and $(p \vee \sim q) \wedge (r \vee F_0)$ are $p \wedge \sim p$, $p \wedge F_0$ and $(p \wedge \sim q) \vee (r \wedge T_0)$ respectively.

Note: $(a^*)^* = a$

The following is an interesting theorem which states that if any two formulas (containing $\wedge, \vee, \sim, T_0$ and F_0) are equivalent then their duals are also equivalent to each other.

Theorem 2: Principle of Duality

Let a and b be formulas (containing $\wedge, \vee, \sim, T_0$ and F_0). If $a \equiv b$ then $a^* \equiv b^*$.

Basic equivalent formulas:

Theorem 3:

If p and q are proposition variables then the following properties hold:

- *Idempotent properties*

$$p \vee p \equiv p \quad ; \quad p \wedge p \equiv p$$

- *Commutative properties*

$$p \vee q \equiv q \vee p \quad ; \quad p \wedge q \equiv q \wedge p$$

- Absorption properties

$$p \wedge (p \vee q) \equiv p ; \quad p \vee (p \wedge q) \equiv p$$

Proof: We prove $p \vee p \equiv p$, $p \vee q \equiv q \vee p$ and $p \vee (p \wedge q) \equiv p$ through truth tables.

p	q	p	$p \vee p$	$p \vee q$	$q \vee p$	$p \wedge (p \vee q)$
T	T	T	T	T	T	T
T	F	T	T	T	T	T
F	T	F	F	T	T	F
F	F	F	F	F	F	F



Note that the truth values of $p \vee p$ and p are identical. Therefore $p \vee p \equiv p$. By the same reason $p \vee q \equiv q \vee p$ and $p \wedge (p \vee q) \equiv p$. The other equivalences follow by the principle of duality.

Theorem 4: Negation properties:

If p and q are proposition variables, then the following properties hold:

- Double negation property

$$\sim(\sim p) \equiv p$$

- De Morgan's properties

$$\sim(p \vee q) \equiv \sim p \wedge \sim q ; \quad \sim(p \wedge q) \equiv \sim p \vee \sim q$$

proof: We prove $\sim(\sim p) \equiv p$ and $\sim(p \vee q) \equiv \sim p \wedge \sim q$ through truth tables.

p	q	$p \vee q$	$\sim(p \vee q)$	$\sim p$	$\sim q$	$\sim p \wedge \sim q$	$\sim(\sim p)$
T	T	T	F	F	F	F	T
T	F	T	F	F	T	F	T
F	T	T	F	T	F	F	F
F	F	F	T	T	T	T	F



Note that the truth values of $\sim(\sim p)$ and p are identical. Therefore $\sim(\sim p) \equiv p$. By the same reason $\sim(p \vee q) \equiv \sim p \wedge \sim q$. The other De Morgan's law follows by the principle of duality.

Theorem 5:

If p, q and r are proposition variables, then the following properties hold:

- Associative properties

$$p \vee (q \vee r) \equiv (p \vee q) \vee r \quad ; \quad p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$$

- Distributive properties

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r) \quad ; \quad p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

Proof: We prove $p \vee (q \vee r) \equiv (p \vee q) \vee r$ and $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ through truth tables.

p	q	r	$(p \vee (q \vee r))$	$((p \vee q) \vee r)$
T	T	T	T	T
T	T	F	T	T
T	F	T	T	T
T	F	F	T	T
F	T	T	T	T
F	T	F	T	T
F	F	T	T	F
F	F	F	F	F
1	1	1	3	1



p	q	r	$(p \wedge (q \vee r))$	$((p \wedge q) \vee (p \wedge r))$
T	T	T	T	T
T	T	F	T	F
T	F	T	T	T
T	F	F	F	F
F	T	T	F	F
F	T	F	F	F
F	F	T	F	F
F	F	F	F	F
1	1	1	3	2



Note that the truth values of $p \vee (q \vee r)$ and $(p \vee q) \vee r$ are identical. Therefore $p \vee (q \vee r) \equiv (p \vee q) \vee r$. By the same reason $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$. The other equivalences follow by the principle of duality.

Theorem 6:

If p is a proposition variable, T_0 and F_0 are special variables denoting a tautology and contradiction respectively, then the following properties hold:

- Identity properties

$$p \wedge T_0 \equiv p \quad ; \quad p \vee F_0 \equiv p$$

- Domination properties

$$p \vee T_0 \equiv T_0 \quad ; \quad p \wedge F_0 \equiv F_0$$

- Inverse properties

$$p \vee \sim p \equiv T_0 \quad ; \quad p \wedge \sim p \equiv F_0$$

Proof: We prove $p \wedge T_0 \equiv p$, $p \vee T_0 \equiv T_0$ and $p \vee \sim p \equiv T_0$ through the truth tables.

p	T_0	F_0		$(p \wedge T_0)$		$(p \vee T_0)$		$(p \vee \sim p)$
T	T	F		T		T		T
F	T	F		F		T		T
1				2		3		4

Note that the truth values of $p \wedge T_0$ and p are identical. Therefore $p \wedge T_0 \equiv p$. By the same reason $p \vee T_0 \equiv T_0$ and $p \vee \sim p \equiv T_0$. The other equivalences follow by the principle of duality.

The equivalences so far given are the properties of the operators $\wedge, \vee$ and $\sim$ on the set of propositions in symbolic logic. The set of all propositions under the operations $\wedge, \vee$ and $\sim$ is an algebra called *algebra of propositions* which is a particular example of a *Boolean algebra*. The following is the list of laws for the algebra of propositions:

The laws of Logic (Logical Equivalences):

If p, q, r are any proposition variables, T_0 is any tautology and F_0 is any contradiction, then the following laws of logic are valid.

Equivalence	Name
$p \vee p \equiv p$; $p \wedge p \equiv p$	Idempotent laws
$p \vee q \equiv q \vee p$; $p \wedge q \equiv q \wedge p$	Commutative laws
$p \wedge (p \vee q) \equiv p$; $p \vee (p \wedge q) \equiv p$	Absorption laws
$\sim(\sim p) \equiv p$	Double negation law
$\sim(p \vee q) \equiv \sim p \wedge \sim q$; $\sim(p \wedge q) \equiv \sim p \vee \sim q$	De Morgan's laws
$p \vee (q \vee r) \equiv (p \vee q) \vee r$ $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$	Associative laws
$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	Distributive laws
$p \wedge T_0 \equiv p$; $p \vee F_0 \equiv p$	Identity laws
$p \vee T_0 \equiv T_0$; $p \wedge F_0 \equiv F_0$	Domination laws
$p \vee \sim p \equiv T_0$; $p \wedge \sim p \equiv F_0$	Inverse laws

Note the logical equivalences in the above table, except the double negation law, are in pairs where each pair of formulas are duals of each other.

Replacement Process:

It is a process in which we replace any part of the *wff* which is itself a formula (be it *atomic* or *molecular*) by any other formula. In general, a replacement yields a new formula.

Replacement rule: Let a be a formula, b be a formula that appears in a , and let c be a formula such that $c \equiv b$. Let a_1 be the formula obtained by replacing one or more occurrences of b by c in a . Then $a_1 \equiv a$.

Further if a is a tautology then a_1 is also a tautology. That is, if we replace any part or parts of a tautology by formulas that are equivalent to these parts we again get a tautology.

Example:

- I. Let a be the formula $(p \rightarrow q) \rightarrow r$. We have $p \rightarrow q \equiv \sim p \vee q$. Let a_1 be the formula obtained by replacing $p \rightarrow q$ by its equivalent formula $\sim p \vee q$ i.e., $a_1: (\sim p \vee q) \rightarrow r$. Then $a \equiv a_1$,
i.e., $(p \rightarrow q) \rightarrow r \equiv (\sim p \vee q) \rightarrow r$

- II. Let b be the formula $p \rightarrow (p \vee q)$. Since $p \equiv \sim(\sim p)$, the formula $b_1: p \rightarrow (\sim(\sim p) \vee q)$ is derived from b by replacing the second occurrence (but not the first occurrence) of p by $\sim(\sim p)$. By replacement rule $b \equiv b_1$

Equivalent formulas for conditional and Biconditional propositions :

Theorem 7: Prove that

$$(a) p \rightarrow q \equiv \sim p \vee q$$

$$(b) p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$(c) p \rightarrow q \equiv \sim q \rightarrow \sim p$$

Solution: We prove (a) and (b) by constructing truth tables

p	q	$p \rightarrow q$	$(\sim p \vee q)$	$p \leftrightarrow q$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	$F \vee T$	T	$T \wedge T$
T	F	F	$F \vee F$	F	$F \wedge F$
F	T	T	$T \vee T$	F	$T \wedge F$
F	F	T	$T \vee T$	T	$T \wedge T$
1	1	2	3 4	5	6 8 7



(c) We have $p \rightarrow q \equiv \sim p \vee q$

$$\text{Now, } \sim q \rightarrow \sim p \equiv \sim(\sim q) \vee \sim p$$

$$\equiv q \vee \sim p \quad (\text{Double negation property, Replacement rule})$$

$$\equiv \sim p \vee q \quad (\text{Commutative law for disjunction})$$

$$\equiv p \rightarrow q$$

$$\text{Therefore } p \rightarrow q \equiv \sim q \rightarrow \sim p \quad (\text{Symmetry of equivalence})$$

Note: A conditional proposition $p \rightarrow q$ and its contrapositive $\sim q \rightarrow \sim p$ are logically equivalent.

Example 1: Show that $(p \rightarrow q) \wedge (\sim q \wedge (r \vee \sim q)) \equiv \sim(p \vee q)$.

Solution: $(p \rightarrow q) \wedge (\sim q \wedge (r \vee \sim q))$

$$\equiv (p \rightarrow q) \wedge (\sim q \wedge (\sim q \vee r)) \quad (\text{Commutative law for } \vee)$$

$$\equiv (p \rightarrow q) \wedge \sim q \quad (\text{Absorption law})$$

$$\equiv (\sim p \vee q) \wedge \sim q \quad (\alpha \rightarrow \beta \equiv \sim \alpha \vee \beta)$$

$$\equiv (\sim p \wedge \sim q) \vee (q \wedge \sim q) \quad (\text{Distributive law})$$

$$\equiv (\sim(p \vee q)) \vee F_0 \quad (\text{De Morgan's law and } q \wedge \sim q \text{ is a contradiction})$$

$$\equiv \sim(p \vee q) \quad (\text{Identity law})$$

Example 2: Show that

$((p \vee q) \wedge \sim(\sim p \wedge (\sim q \vee \sim r))) \vee (\sim p \wedge \sim q) \vee (\sim p \wedge \sim r)$ is a tautology.

Solution:

$$(\sim p \wedge \sim q) \vee (\sim p \wedge \sim r) \equiv \sim(p \vee q) \vee \sim(p \vee r) \quad (\text{De Morgan's law})$$

$$\equiv \sim((p \vee q) \wedge (p \vee r)) \quad (\text{De Morgan's law})$$

$$\equiv \sim(p \vee (q \wedge r)) \quad (\text{Distributive law})$$

and

$$(p \vee q) \wedge \sim(\sim p \wedge (\sim q \vee \sim r))$$

$$\equiv (p \vee q) \wedge \sim(\sim p \wedge \sim(q \wedge r)) \quad (\text{De Morgan's law})$$

$$\equiv (p \vee q) \wedge \sim(\sim(p \vee (q \wedge r))) \quad (\text{De Morgan's law})$$

$$\equiv (p \vee q) \wedge ((p \vee (q \wedge r))) \quad (\text{Double negation law})$$

$$\equiv (p \vee q) \wedge ((p \vee q) \wedge (p \vee r)) \quad (\text{Distributive law})$$

$$\equiv ((p \vee q) \wedge (p \vee q)) \wedge (p \vee r) \quad (\text{Associative law})$$

$$\equiv (p \vee q) \wedge (p \vee r) \equiv p \vee (q \wedge r) \quad (\text{Idempotent law})$$

Now, the given formula is equivalent to $p \vee (q \wedge r) \vee \sim(p \vee (q \wedge r)) \equiv T_0$
 Since it is a substitution instance of a tautology $p \vee \sim p$ (by substituting p by $p \vee (q \wedge r)$).

Example 3: Show that

$$(i) \quad \sim(p \wedge q) \rightarrow (\sim p \vee (\sim p \vee q)) \equiv \sim p \vee q$$

$$(ii) \quad (p \vee q) \wedge (\sim p \wedge (\sim p \wedge q)) \equiv \sim p \wedge q$$

Solution: $\sim(p \wedge q) \rightarrow (\sim p \vee (\sim p \vee q))$

$$\equiv \sim(\sim(p \wedge q)) \vee (\sim p \vee (\sim p \vee q)) \quad (\alpha \rightarrow \beta \equiv \sim \alpha \vee \beta)$$

$$\equiv (p \wedge q) \vee ((\sim p \vee \sim p) \vee q) \quad (\text{Double negation law and Associative law for } \vee)$$

$$\equiv (p \wedge q) \vee (\sim p \vee q) \quad (\text{Idempotent law})$$

$$\equiv ((p \wedge q) \vee \sim p) \vee q \quad (\text{Associative law for } \vee)$$

$$\equiv ((p \vee \sim p) \wedge (q \vee \sim p)) \vee q \quad (\text{Distributive law})$$

$$\equiv (T_0 \wedge (q \vee \sim p)) \vee q \quad (\text{Inverse law})$$

$$\equiv (q \vee \sim p) \vee q \quad (\text{Identity law})$$

$$\equiv (\sim p \vee q) \vee q \quad (\text{Commutative law for } \vee)$$

$$\equiv \sim p \vee (q \vee q) \quad (\text{Associative law for } \vee)$$

$$\equiv \sim p \vee q \quad (\text{Idempotent law})$$

This proves (i). Note that

$$\sim(p \wedge q) \rightarrow (\sim p \vee (\sim p \vee q)) \equiv (p \wedge q) \vee (\sim p \vee (\sim p \vee q)) \equiv \sim p \vee q$$

$$\text{Therefore } (p \wedge q) \vee (\sim p \vee (\sim p \vee q)) \equiv \sim p \vee q$$

By the principle of duality

$$(p \vee q) \wedge (\sim p \wedge (\sim p \wedge q)) \equiv \sim p \wedge q$$

P1:

Show that $\sim(p \vee (\sim p \wedge q))$ and $\sim p \wedge \sim q$ are logically equivalent.

Solution:

We prove this by using logical equivalences.

$$\begin{aligned}\sim(p \vee (\sim p \wedge q)) &\equiv \sim((p \vee \sim p) \wedge (p \vee q)) && \text{(Distributive law)} \\ &\equiv \sim(T_0 \wedge (p \vee q)) && (\because p \vee \sim p \text{ is a tautology}) \\ &\equiv \sim(p \vee q) && \text{(Identity law)} \\ &\equiv \sim p \wedge \sim q && \text{(De Morgan's law)}\end{aligned}$$

P2:

Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

Solution:

We show this by using logical equivalences.

$(p \wedge q) \rightarrow (p \vee q) \equiv \sim(p \wedge q) \vee (p \vee q)$	$(\alpha \rightarrow \beta \equiv \sim\alpha \vee \beta)$
$\equiv (\sim p \vee \sim q) \vee (p \vee q)$	(De Morgans' law)
$\equiv ((\sim p \vee \sim q) \vee p) \vee q$	(Associative law for $\vee$)
$\equiv (\sim p \vee (\sim q \vee p)) \vee q$	(Associative law for $\vee$)
$\equiv (\sim p \vee (p \vee \sim q)) \vee q$	(Commutativity for $\vee$)
$\equiv ((\sim p \vee p) \vee \sim q) \vee q$	(Associative law for $\vee$)
$\equiv (\sim p \vee p) \vee (\sim q \vee q)$	(Associative law for $\vee$)
$\equiv T_0 \vee T_0 = T_0$	

P3:

Show that $\sim(\sim((p \vee q) \wedge r) \vee \sim q)$ and $q \wedge r$ are logical equivalent.

Solution:

We prove this by using logical equivalences.

$$\begin{aligned}\sim(\sim((p \vee q) \wedge r) \vee \sim q) &\equiv \sim(\sim((p \vee q) \wedge r)) \wedge \sim(\sim q) && \text{(De Morgan's law)} \\ &\equiv ((p \vee q) \wedge r) \wedge q && \text{(Double negation law)} \\ &\equiv q \wedge ((p \vee q) \wedge r) && \text{(Commutative law for } \wedge \text{)} \\ &\equiv (q \wedge (p \vee q)) \wedge r && \text{(Associative law for } \wedge \text{)} \\ &\equiv (q \wedge (q \vee p)) \wedge r && \text{(Commutative law for } \vee \text{)} \\ &\equiv q \wedge r && \text{(Absorption law)}\end{aligned}$$

P4:

Prove the following logical equivalence

$$p \wedge \left((\sim q \rightarrow (r \wedge r)) \vee \sim (q \vee ((r \wedge s) \vee (r \wedge \sim s))) \right) \Leftrightarrow p$$

Solution:

$$p \wedge \left((\sim q \rightarrow (r \wedge r)) \vee \sim (q \vee ((r \wedge s) \vee (r \wedge \sim s))) \right)$$

$$\equiv p \wedge \left((\sim q \rightarrow r) \vee \sim (q \vee (r \wedge (s \vee \sim s))) \right)$$

(Idempotent law and Distributive law)

$$\equiv p \wedge \left((\sim(\sim q) \vee r) \vee \sim (q \vee (r \wedge T_0)) \right)$$

($\alpha \rightarrow \beta \equiv \sim \alpha \vee \beta$, inverse law)

$$\equiv p \wedge \left((q \vee r) \vee \sim (q \vee r) \right)$$

(Double negation law, identity law)

$$\equiv p \wedge T_0$$

(Substitution instance of a tautology)

$$\equiv p$$

(Identity law)

P5:

Show that $(\sim p \wedge (\sim q \wedge r)) \vee (q \wedge r) \vee (p \wedge r) \Leftrightarrow r$

Solution:

$$(\sim p \wedge (\sim q \wedge r)) \vee (q \wedge r) \vee (p \wedge r)$$

$$\Leftrightarrow (\sim p \wedge (\sim q \wedge r)) \vee ((q \vee p) \wedge r) \quad (\text{Distributive law})$$

$$\Leftrightarrow ((\sim p \wedge \sim q) \wedge r) \vee ((q \vee p) \wedge r) \quad (\text{Associative law for } \wedge)$$

$$\Leftrightarrow ((\sim p \wedge \sim q) \vee (q \vee p)) \wedge r \quad (\text{Distributive law})$$

$$\Leftrightarrow (\sim(p \vee q) \vee (p \vee q)) \wedge r \quad (\text{De Morgan's law and commutativity of } \vee)$$

$$\Leftrightarrow T_0 \wedge r \quad (\text{Substitution instance of a tautology and commutativity of } \vee)$$

$$\Leftrightarrow r \quad (\text{Identity law})$$

P6:

Prove the following without using truth tables

$$(a) \sim(p \rightarrow q) \equiv p \wedge \sim q$$

$$(b) \sim(p \leftrightarrow q) \equiv (p \wedge \sim q) \vee (\sim p \wedge q)$$

Solution:

$$\begin{aligned} (a) \sim(p \rightarrow q) &\equiv \sim(\sim p \vee q) && (\because p \rightarrow q \equiv \sim p \vee q) \\ &\equiv (\sim(\sim p)) \wedge \sim q && (\text{De Morgan's law}) \\ &\equiv p \wedge \sim q && (\text{Double negation law}) \end{aligned}$$

$$\begin{aligned} (b) \sim(p \leftrightarrow q) &\equiv \sim((p \rightarrow q) \wedge (q \rightarrow p)) \\ &\equiv \sim((\sim p \vee q) \wedge (\sim q \vee p)) \\ &\equiv (\sim(\sim p \vee q)) \vee (\sim(\sim q \vee p)) && (\text{De Morgan's law}) \\ &\equiv ((\sim(\sim p) \wedge \sim q)) \vee ((\sim(\sim q) \wedge \sim p)) && (\text{De Morgan's law}) \\ &\equiv (p \wedge \sim q) \vee (q \wedge \sim p) && (\text{Double negation law}) \\ &\equiv (p \wedge \sim q) \vee (\sim p \wedge q) && (\text{Commutative law for conjunction}) \end{aligned}$$

P7:

Show that

$$(a) (p \wedge q) \rightarrow r \equiv p \rightarrow (q \rightarrow r)$$

$$(b) (p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$$

Solution:

$$\begin{aligned} (a) (p \wedge q) \rightarrow r &\equiv (\sim(p \wedge q)) \vee r && (\alpha \rightarrow \beta \equiv \sim\alpha \vee \beta) \\ &\equiv (\sim p \vee \sim q) \vee r && \text{(De Morgan's law)} \\ &\equiv \sim p \vee (\sim q \vee r) && \text{(Associative law for } \vee) \\ &\equiv p \rightarrow (\sim q \vee r) && (\alpha \rightarrow \beta \equiv \sim\alpha \vee \beta) \\ &\equiv p \rightarrow (q \rightarrow r) && (\alpha \rightarrow \beta \equiv \sim\alpha \vee \beta) \end{aligned}$$

$$\begin{aligned} (b) (p \vee q) \rightarrow r &\equiv \sim(p \vee q) \vee r && (\alpha \rightarrow \beta \equiv \sim\alpha \vee \beta) \\ &\equiv (\sim p \wedge \sim q) \vee r && \text{(De Morgan's law)} \\ &\equiv (\sim p \vee r) \wedge (\sim q \vee r) && \text{(Distributive law)} \\ &\equiv (p \rightarrow r) \wedge (q \rightarrow r) && (\alpha \rightarrow \beta \equiv \sim\alpha \vee \beta) \end{aligned}$$

P8:

There are two restaurants next to each other. One has a sign that says, “*Good food is not cheap*”, and the other has a sign that says, “*Cheap food is not good*”. Are the signs saying the same thing?

Solution:

Let g and c respectively denote the propositions *The food is good* and *The food is cheap*.

The first sign *Good food is not cheap* can be written as $g \rightarrow \sim c$.

The second sign *Cheap food is not good* can be written as $c \rightarrow \sim g$.

Now, $g \rightarrow \sim c \equiv \sim g \vee \sim c$ and $c \rightarrow \sim g \equiv \sim c \vee \sim g$.

Then $g \rightarrow \sim c \equiv \sim g \vee \sim c \equiv \sim c \vee g \equiv c \rightarrow \sim g$.

$$\therefore g \rightarrow \sim c \equiv c \rightarrow \sim g$$

Thus, the two signs are equivalent. Therefore, they say the same thing.

1.3. Equivalence Formulas, Logical Equivalence and the Laws of Logic

Exercises:

I. Show the following equivalences:

$$1. (p \vee q) \wedge \sim(\sim p \wedge q) \equiv p$$

$$2. p \vee q \vee (\sim p \wedge (\sim q \wedge r)) \Leftrightarrow p \vee q \vee r$$

$$3. \sim(p \vee q) \vee ((\sim p \wedge q) \vee \sim q) \Leftrightarrow \sim(p \wedge q)$$

$$4. p \rightarrow (q \vee r) \equiv (p \rightarrow q) \vee (p \rightarrow r)$$

$$5. \sim(p \leftrightarrow q) \equiv (p \vee q) \wedge \sim(p \wedge q)$$

$$6. ((\sim p \vee \sim q) \rightarrow (p \wedge q \wedge r)) \Leftrightarrow p \wedge q$$

$$7. ((p \wedge q) \vee (p \wedge r)) \rightarrow s \equiv (\sim p \vee (\sim q \wedge \sim r)) \vee s$$

II. Show that the following are tautologies.

$$1. (p \vee q) \rightarrow (q \rightarrow q)$$

$$2. p \rightarrow (q \rightarrow (p \wedge q))$$

$$3. ((p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)) \rightarrow r$$